

EduQuest

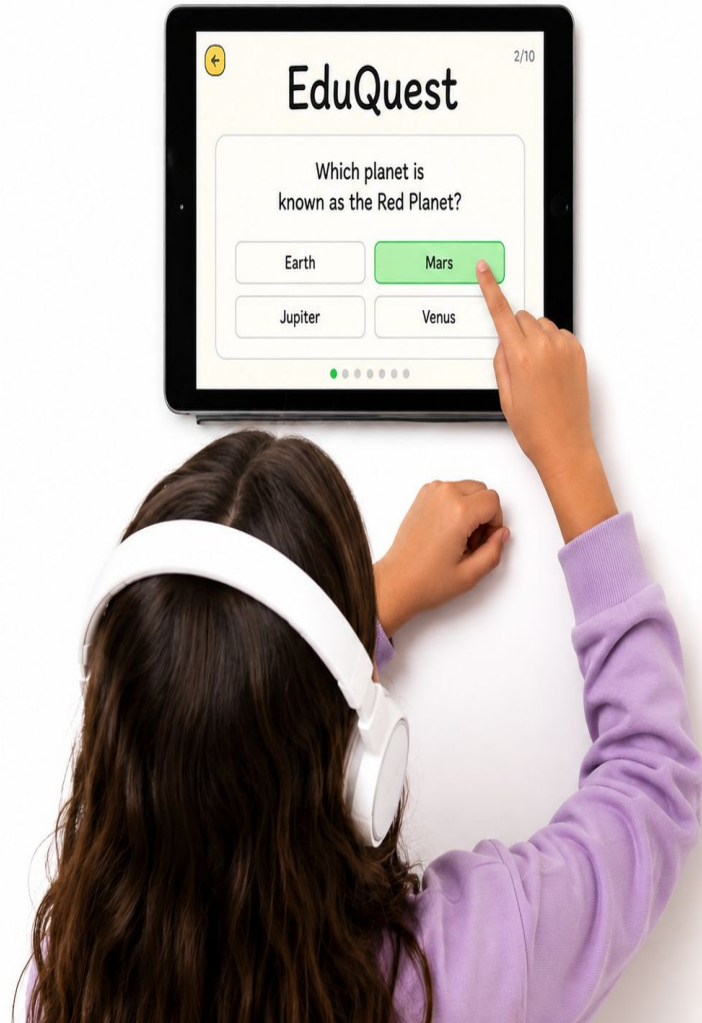
TEAM DIAMOND

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CS 410 - PROFESSIONAL WORKFORCE DEVELOPMENT I

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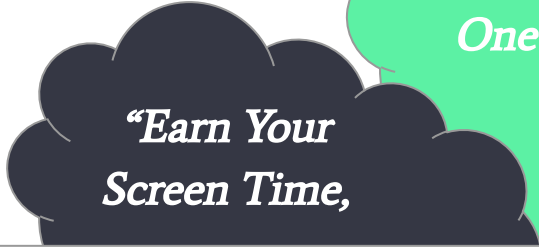
Prototype Disclaimer

As is stand, we cannot enforce OS-level enforcement for the EduQuest. In order to do so we would need to submit an application to Apple and Android and have them review EduQuest for it get permission. We do not have the time or resources to do that. For this product we will make it without the OS-level enforcement with the idea of expanding upon EduQuest's capabilities. Without the OS-level enforcement, we cannot prevent children from uninstalling EduQuest. Nor can prevent kids from using devices with a set limit. EduQuest will operate as a standalone app that the children will have to interact with willingly and to do questions on their own. This presentation is made based on the idea that we DO have permission and the capability to use OS-level enforcement.

ELEVATOR PITCH

The Problem

Children spend a lot of time using entertainment screens, displacing time that could be used to reinforce academic skills.



*"Earn Your
Screen Time,"*



*One Question at
a Time"*

The Solution

EduQuest turns recreational screen time into an opportunity for learning by requiring children to answer short educational questions before entertainment apps unlock.

How It Works

Fixed Screen Time → Questions Mode → Answer Questions → Earn More Screen Time

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PROBLEM STATEMENT

Reading and math scores among 9-year-old students have **dropped** to their lowest levels in decades .

At the same time, children ages 8 to 12 average approximately 5.5 hours of recreational screen time each day, it **displaces** valuable time that could be otherwise be used to reinforce foundation skills

¹ National Center for Education Statistics (See References A)

² Rideout, Victoria, et al. (See References A)



PROBLEM CHARACTERISTICS

The challenge is not simply how much screen time children have, but how that time is being used.



Entertainment Dominates

> 4 hours daily video/gaming vs. 9 minutes e-reading.¹



Parents Struggle

47% of parents struggle to manage screen time for their children.²

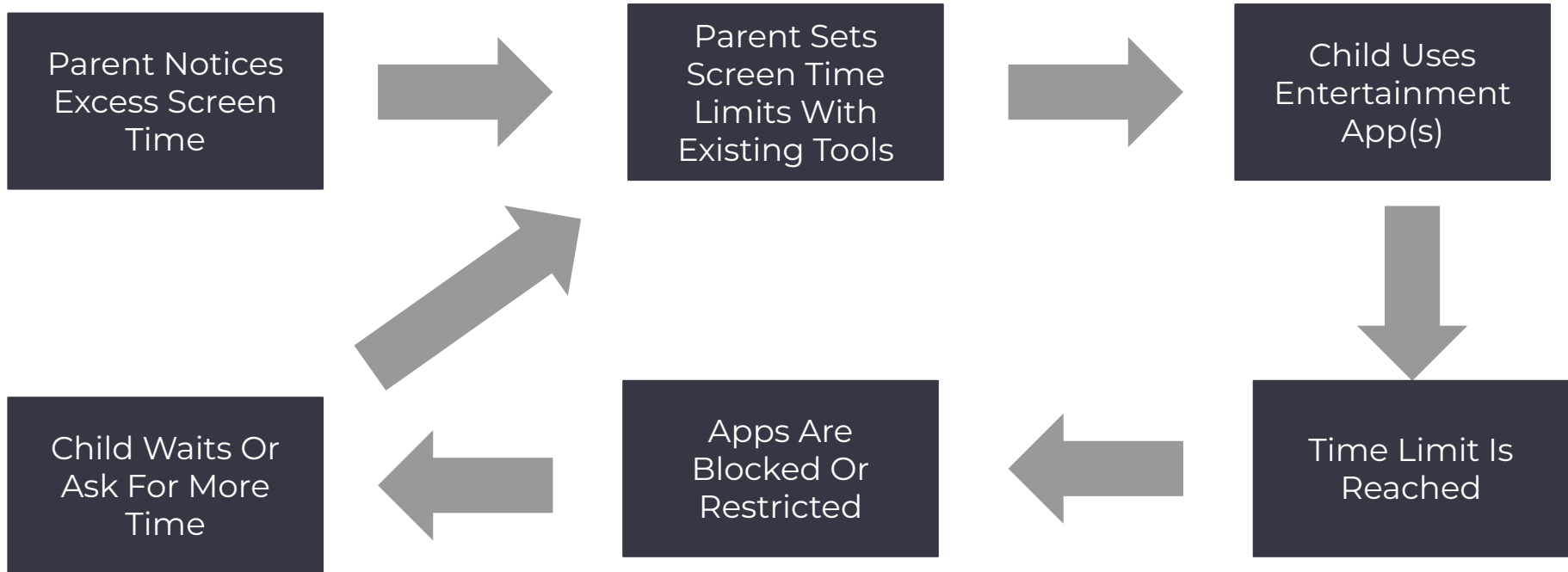


Inconsistent Enforcement

Only 19% of parents consistently follow their children's screen time rules.²

¹ Rideout et al. (See Reference A) | ² Pew Research. (See Reference G)

CURRENT PROCESS FLOW



THE EDUCATIONAL GAP

The current process simply restricts time usage, resulting in a missed opportunity for learning and zero educational reinforcement.

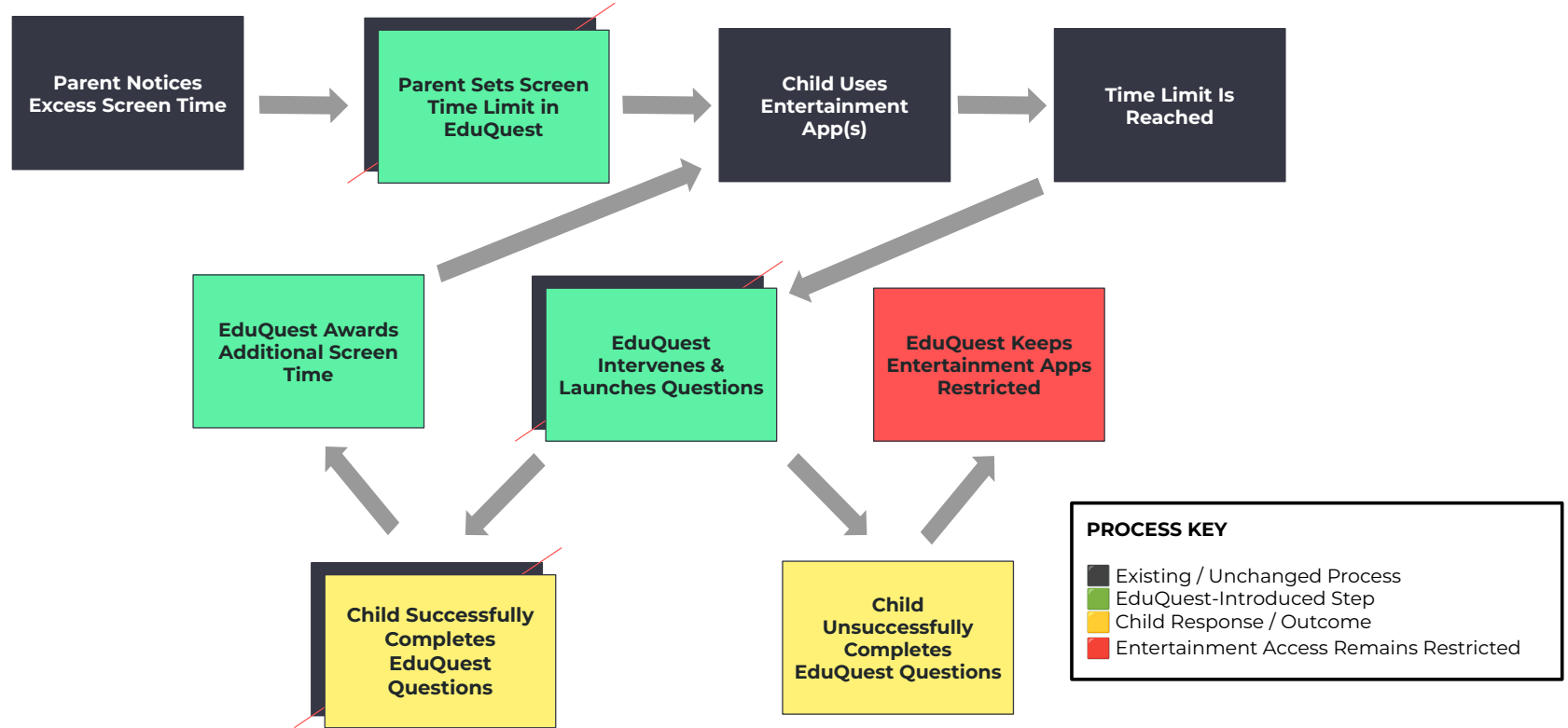
SOLUTION STATEMENT

EduQuest is a mobile application controlled by parents that combines **screen time management with educational practice.**

The application allows parents to set screen-time limits and **requires children to complete age appropriate questions** to earn additional screen time. Parents are also able to monitor learning progress over time.



SOLUTION PROCESS FLOW



WHAT EDUQUEST **WILL** DO

Set

Parents set recreational screen-time limits.

Learn

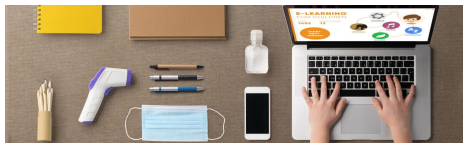
Children answer age-appropriate reading and math questions.

Earn

Completed questions earn additional recreational screen time.

Track

Parents monitor question performance, screen-time activity, and learning progress.



WHAT EDUQUEST WILL **NOT** DO

Replace Education

EduQuest supports practice and reinforcement, but it will not replace teachers, schools, or formal instruction.

Eliminate Screen Time

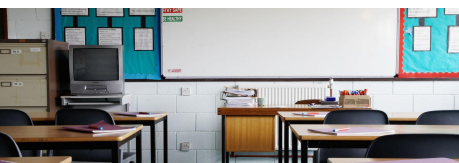
EduQuest will not completely block recreational screen use. It is designed to encourage a balance between learning and entertainment.

Replace Parental Control

Parents will still decide appropriate screen-time limits and manage their child's settings.

Guarantee Results

EduQuest can encourage educational practice, but it cannot guarantee academic improvement.



COMPETITION MATRIX

Sources: Qustodio; Bark Technologies; Apple; Khan Academy. (References C)

Features	EduQuest	Qustodio (Direct)	Bark (Direct)	Apple Screen Time (Direct)	Khan Academy Kids (Indirect)
Earning Screen Time Through Academic Activity	✓				
Academic Content	✓				✓
Direct Parental Control	✓	✓	✓	✓	
Parent Dashboard Monitoring	✓				*
Content Filtering		✓	✓	✓	
Cross-platform	✓	✓	✓	*	✓
Screen-Locking	✓				

Slide Credit: Ethan Lemerande

Legend: ✓ Feature Supported * Partially supported □ Not supported

DEVELOPMENT TOOLS

IDE

Visual Studio Code

Used to write, edit, and manage code.

Version Control

Git & GitHub

Used to manage source code, track changes, and collaborate as a team.

CI/CD

GitHub Actions & Workflows

Used to automate project builds, testing, and development workflows.

Additional Tools

To Be Determined

Backend, frontend, testing, and documentation tools will be selected later.

MAJOR FUNCTIONAL COMPONENTS

OS: Linux for server's operating system wrapped in Docker

Server framework: FastAPI or Django

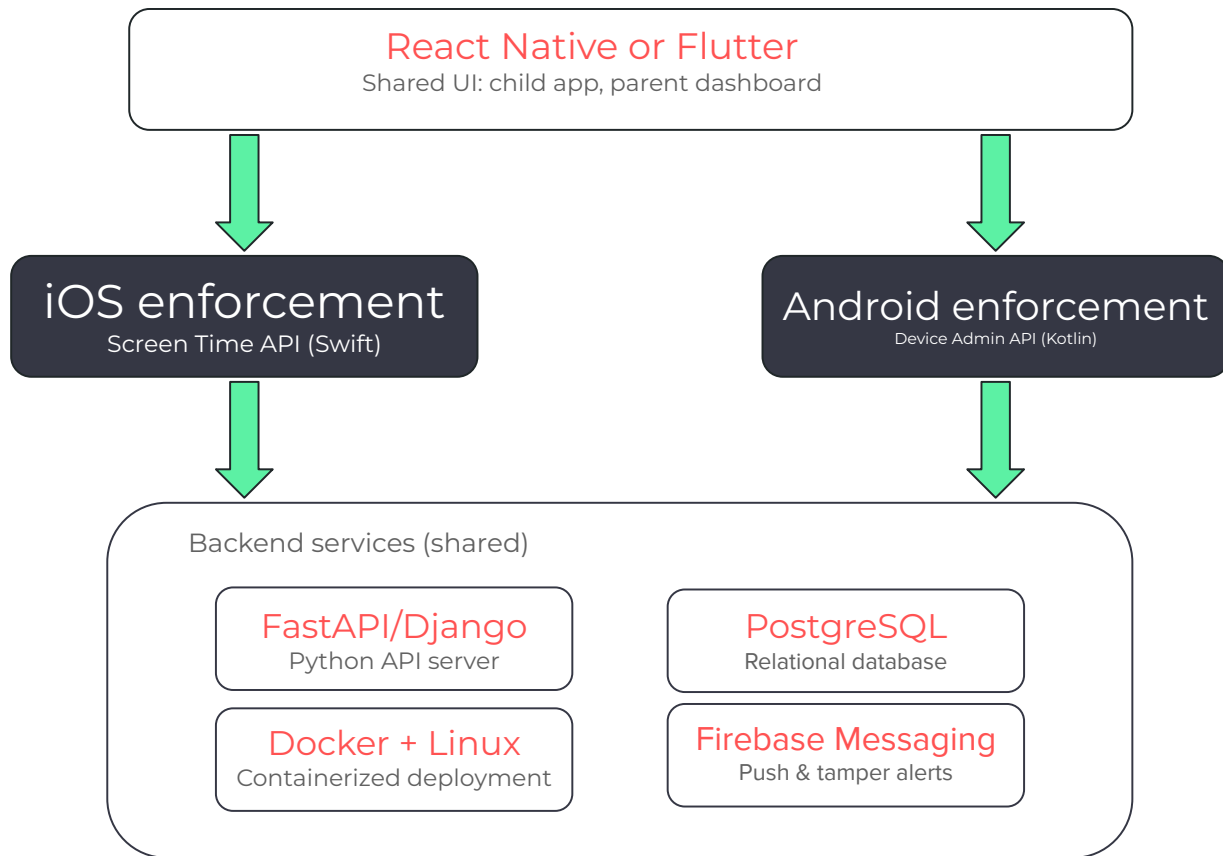
Database: PostgreSQL

Language: Python

Push notifications: Firebase Cloud Messaging

Mobile frontend: React Native or Flutter

MAJOR FUNCTIONAL COMPONENTS DIAGRAM



RISKS

Risk	Probability and Impact	Mitigation	After Mitigation
Customer & End User: Children may become frustrated with the parental controls and stop engaging with the app altogether. Parents could get frustrated with all of the parental control options.	P: 4/5 I: 4/5	Rewards for engaging and completing the required sections, adaptive learning methods for less frustration, screen-time limits, and instructions for how to navigate the parental controls and what each control does specified in detail.	P: 2/5 I: 2/5
Technical: Children may attempt to uninstall the app or bypass the parental controls.	P: 4/5 I: 5/5	Regularly test bypass methods, push notifications for deletion requests, and authentication for the app being uninstalled.	P: 1/5 I: 2/5

RISKS

Risk	Probability and Impact	Mitigation	After Mitigation
Security: Unauthorized access could expose sensitive account information, learning results and progress, and screen-time settings.	P: 3/5 I: 5/5	Secure authorization methods, encryption, limited data access, and regular security testing and updates.	P: 1/5 I: 3/5
Legal: This platform caters to school-aged children, so privacy and compliance concerns arise because of data handling and collection (Federal Trade Commission).	P: 2/5 I: 5/5	Parental authorization, minimal data collection, clear privacy practices and policies, and parental controls over every child account (Children's Online Privacy Protection Rule).	P: 1/5 I: 3/5

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APPENDIX

Appendix A: Key Terms & Concepts

Recreational Screen Time

Time spent using a device for fun, such as playing games, watching videos, or using social media.

Screen-Time Management

Tools used to track or limit how much time someone spends on a device or app.

Parent Dashboard

A screen where parents can view and manage their child's activity and learning progress.

Adaptive Learning

A learning method that changes based on how well a student is doing.

Content Filtering

Tools that block or limit access to certain apps, websites, or other digital content.

Cross-Platform

Software that can work on more than one type of device or operating system.

Ecosystem-Locked

Software that only works within one company's devices or services.

Appendix B: Basic Technical Terms

Frontend

The part of an app that users see and interact with.

Backend

The part of an app that works behind the scenes to process information and make features work.

Relational Database

A database that stores information in tables that can be connected to each other.

Server Framework

A set of tools developers use to build the part of an app that runs on a server.

API (Application Programming Interface)

A tool that allows different programs or parts of an app to communicate with each other.

Screen Time API

Apple tools that allow an app to work with screen-time controls on Apple devices.

Device Administration API

Android tools that allow an app to manage certain settings and controls on a device.

APPENDIX

Appendix C: Development Technologies

Linux

An operating system often used to run servers and online services.

Python

A programming language that can be used to build the behind-the-scenes parts of an app.

Swift

A programming language used to build software for Apple devices.

Kotlin

A programming language commonly used to build Android apps.

PostgreSQL

A database system used to store and organize information.

FastAPI

A Python tool used to build the backend of an application.

Django

A Python tool used to build websites and the backend of applications.

React Native

A tool that lets developers build mobile apps for both iOS and Android.

Flutter

A tool that lets developers build apps for different types of devices using the same code.

Docker

A tool that packages an application with the things it needs to run.

Container

A package that holds an application and the files and tools it needs to run.

Containerized Deployment

Running an application using containers so it works the same way in different environments.

Firebase Cloud Messaging

A service used to send notifications and messages to mobile devices.

APPENDIX

Appendix D: Development Process

IDE (Integrated Development Environment)

A program developers use to write, edit, and work with computer code.

Version Control

A system that keeps track of changes made to computer code over time.

Git

A version-control tool used to track changes made to code.

GitHub

An online service where teams can store code and work on it together.

CI/CD (Continuous Integration/Continuous Delivery)

A process that helps teams automatically test, build, and prepare new code for release.

GitHub Actions

A GitHub tool that can automatically run tasks such as testing or building code.

Workflow

A set of steps or tasks that are completed in a certain order.

Appendix E: Security & Data Terms

Authentication

Checking that someone really is who they say they are.

Authorization

Deciding what a user is allowed to see or do after they sign in.

Encryption

Protecting information by changing it into a form that unauthorized people cannot easily read.

Data Access

The ability to view, use, or change stored information.

Data Collection

Gathering and storing information from or about users.

Parental Authorization

Permission from a parent or guardian for a child's account or use of certain features.